Math Class 9th Federal Board Chapter 06 Trigonometry and Bearing Part-I

Trigonometry	**[SLO: M-09-B-20]:** Identify angles in standard position, expressed in degrees and radians.	Summative	K
	[SLO: M-09-B-21]: Apply Pythagoras' theorem and the sine, cosine and tangent ratios for acute angles to find a side or of an angle of a right-angled triangle.	Summative	A
	[SLO: M-09-B-22]: Solve real life trigonometric problems in two dimensions involving angles of elevation and depression.	Summative	A
Trigonometric Identities	**[SLO: M-09-B-23]:** Prove the trigonometric identities and apply them to show different trigonometric relations.	Summative	K/A
	[SLO: M-09-B-24]: Solve real life problems involving trigonometric identities.	Summative	A
Bearing	**[SLO: M-09-B-25]:** Interpret and use three figure bearings.	Summative	K
	[SLO: M-09-B-26]: Solve problems involving bearing.	Summative	U
	[SLO: M-09-B-27]: Apply the concepts of trigonometry	Summative	A

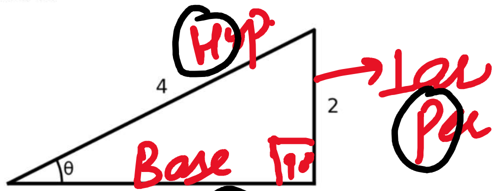
Rule To Gain Full Marks.

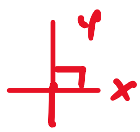
Theory + Examples + Exercise Practice + Model Paper + Past papers (2026, 2025)

1. From the given right-angled triangle, find θ.

$\sin\theta = \frac{2}{4} = \frac{1}{2}$

$\theta = \sin^{-1}\left(\frac{1}{2}\right)$





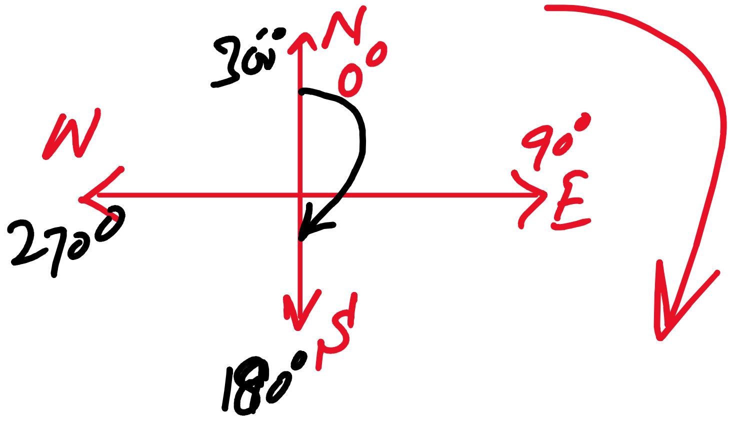
A) 90°	B) 60°	C) 30°	D) 0°

Some people have Curly Brown hair Through proper brushing hair through proper bracketing

$$\boxed{\sin\theta = \frac{P}{H}, \quad \cos\theta = \frac{B}{H}, \quad \tan\theta = \frac{P}{B}}$$

2. If a plane flies on a bearing of 180°, it is moving towards:

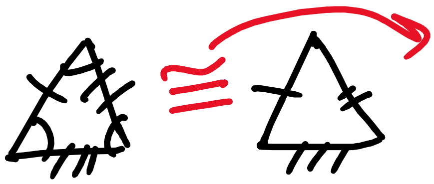
A) North	B) South	C) East	D) West



Clockwise

3. If two triangles are equal, then their corresponding sides are:

A) Equal	B) Unequal	C) Unproportional	D) Perpendicular

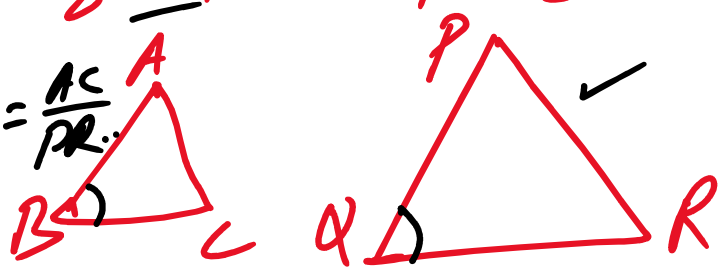


Congruent Triangles

Equal

Type-2 Similar Triangles

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$$



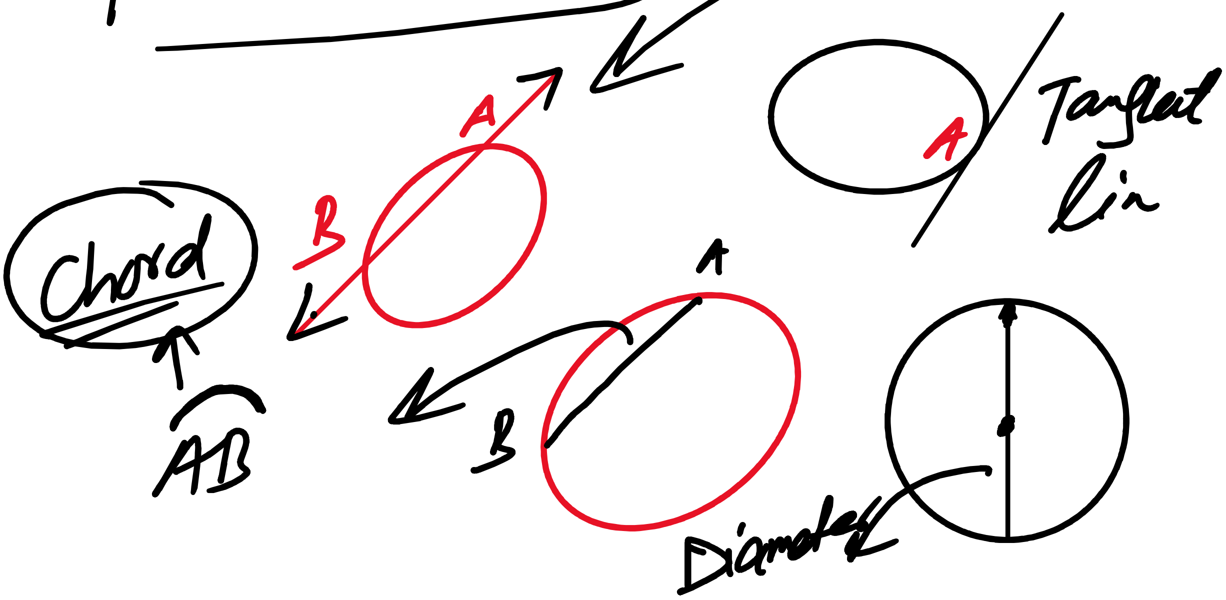
$\angle B = \angle Q, \quad \angle A = \angle P, \quad \angle C = \angle R$

4. 1/sin θ =

A) tan θ	B) cos θ	C) sec θ	D) cosec θ

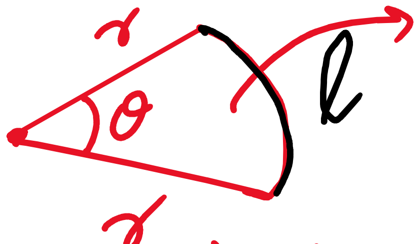
$\sin\theta = \frac{P}{H}$, $\frac{1}{\sin\theta} = \frac{H}{P} = \underline{\underline{\operatorname{cosec}\theta}}$

$\boxed{\sec\theta = \frac{1}{\cos\theta}}$ Secant



Area of Seeter

$= \frac{1}{2} r^2 \theta$



'θ' in radian

$l = r\theta$, $\boxed{A = \frac{1}{2} r^2 \theta}$

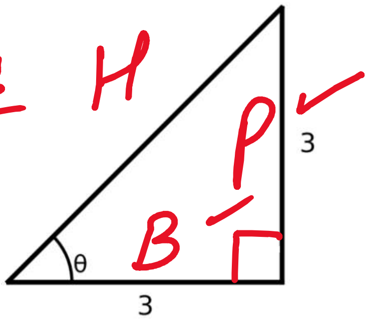
Radius

Radius

$$\boxed{\cot\theta = \frac{1}{\tan\theta}}$$

5. In the given right-angled triangle, find θ.

$\tan\theta = \frac{P}{B} = \frac{\cancel{3}}{\cancel{3}}$ $\theta = \tan^{-1}(1)$

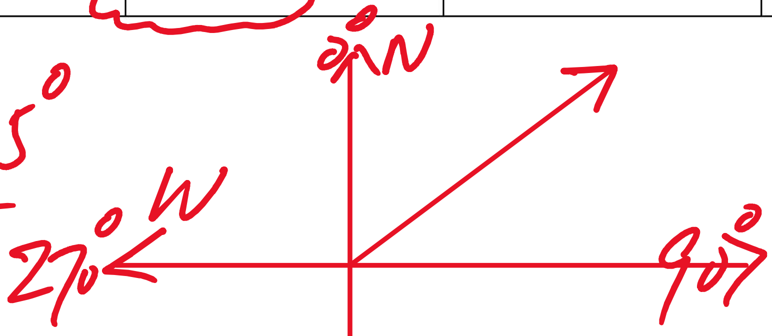


A) 45°	B) 60°	C) 90°	D) 120°

6. A cyclist travels from P to Q on a bearing of 45°. This means the cyclist moves:

A) South-East	B) North-East	C) South-West	D) North-West

$\underline{045^\circ}$



N 45° E

7. $1 + \tan^2\theta =$

A) $\sin^2\theta$	B) $\cos^2\theta$	C) $\sec^2\theta$	D) $\operatorname{cosec}^2\theta$

i. $\sin^2\theta + \cos^2\theta = 1$
ii. $\sec^2\theta - \tan^2\theta = 1$
iii. $\mathrm{cosec}^2\theta - \cot^2\theta = 1$

Sec, ta
Cose, cot

$\sec^2\theta - \tan^2\theta = 1$

$$\boxed{\begin{aligned} 1+\tan^2\theta &= \sec^2\theta \\ 1+\cot^2\theta &= \mathrm{cosec}^2\theta \end{aligned}}$$

$\sin^2\theta + \cos^2\theta = 1$

$\sin^2\theta = ?\quad 1-\cos^2\theta$

$\cos^2\theta = 1-\sin^2\theta$

$= 0.2763$

8. What is the radian measure of 15°50'?

A) 19π/216	B) 19π/36	C) 19π/180	D) 216π/19

DMS'

° → Degree
' → Minute

$1^\circ = 60'$

$60 \times 1'$

$60' = 1^\circ$

$^\circ \to$

$' \to$ Minute

$'' \to$ second

$60' = 1$

$$1' = \left(\frac{1}{60}\right)^\circ$$

$$1' = 60''$$

$$1'' = \left(\frac{1}{60}\right)'$$

$$1^\circ = 60' = 60 \times 1' = 60 \times 60''$$

$$1^\circ = 3600''$$

$50 \times 1'$

10 min

$\Rightarrow \left(\frac{10}{60}\right) hr$

$15^\circ 50'$

$$= 15 + 50'$$

$$= 15^\circ + \left(\frac{5\cancel{0}}{6\cancel{0}}\right)^\circ = \left(15 + \frac{5}{6}\right)^\circ = \left(\frac{35}{6}\right)^\circ$$

$$15^\circ + 0.83 = 15.83$$

$$15^\circ + 0.83 = 15.83$$

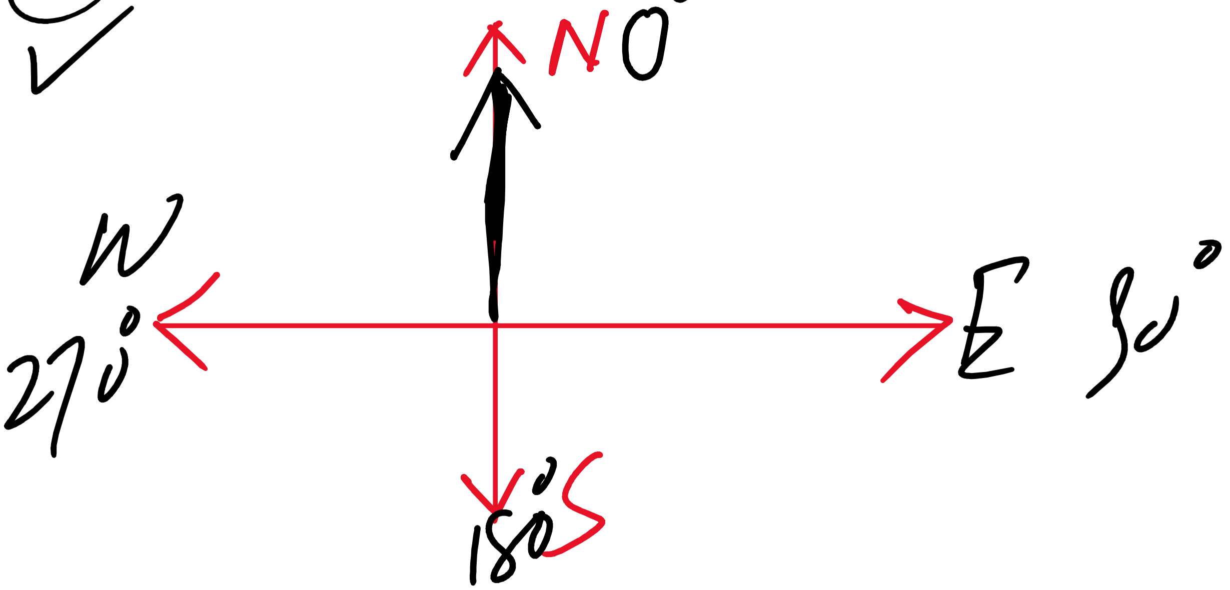
$$15^\circ 50' = \left(\frac{95}{6}\right)^\circ$$

$$= \frac{95}{6} \times \frac{\pi}{180}$$

$$= \frac{18\pi}{216}$$

9. If a navigator gives bearing 0°, in which direction should he travel?

A) North 0°	B) South	C) East	D) West



10. In the figure, if α = β, what is the value of b? → Similar figure

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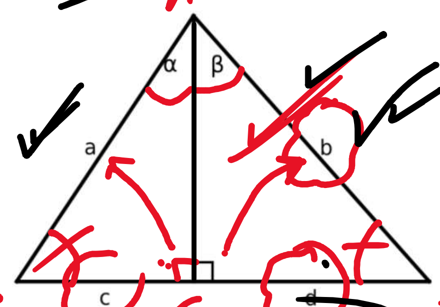
similar figure → Triangle



$\frac{b}{a} = \frac{d}{c}$

A) cd/a	B) c/(ad)	C) ad/c	D) ac/d

$b = \frac{ad}{c}$

$\frac{c}{d} \times \frac{a}{b}$

$bc = ad$

$b = \frac{ad}{c}$

45

11. What is the value of $-3 - 3\tan^2\theta$ in a single trigonometric function?

A) $3\text{cosec}^2\theta$	B) $-3\sec^2\theta$	C) $3\sec^2\theta$	D) $-3\text{cosec}^2\theta$

$-3(1 + \tan^2\theta)$

$= -3\sec^2\theta$

$\text{cosec}\,\theta = \frac{1}{\sin\theta}$

$\cot\theta = \frac{1}{\tan\theta}$

$-2 - 2\cot^2\theta$

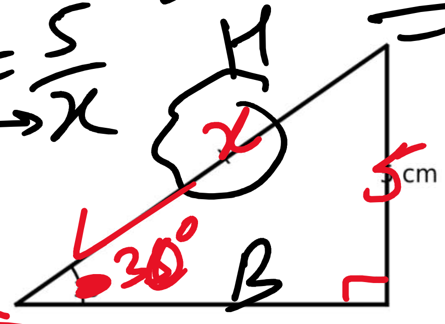
$= -2(1 + \cot^2\theta)$

$= -2\,\text{cosec}^2\theta$

$= -2\ \text{cosec}$

12. What is the value of x in the given figure?

$\sin 35^\circ = \frac{P}{H} = \frac{5}{x}$



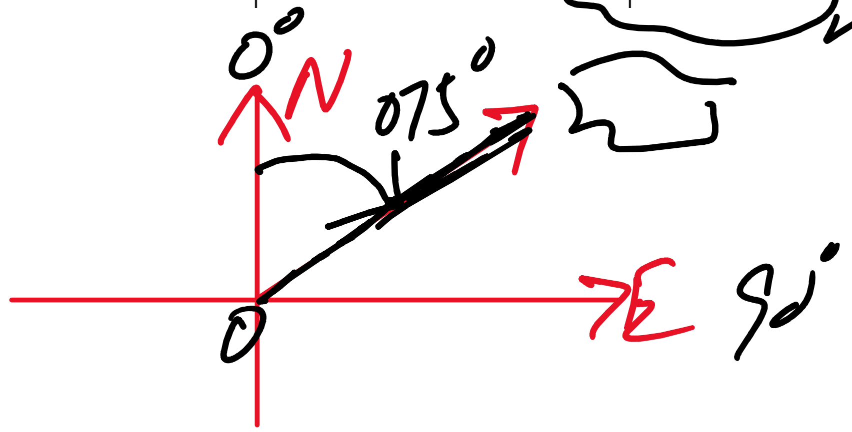
$\Rightarrow x = \frac{5}{\sin 35^\circ} = 8.--$

A) 5 cm	B) 10 cm	C) 7.5 cm	D) 15 cm

$x \sin 35^\circ = 5$

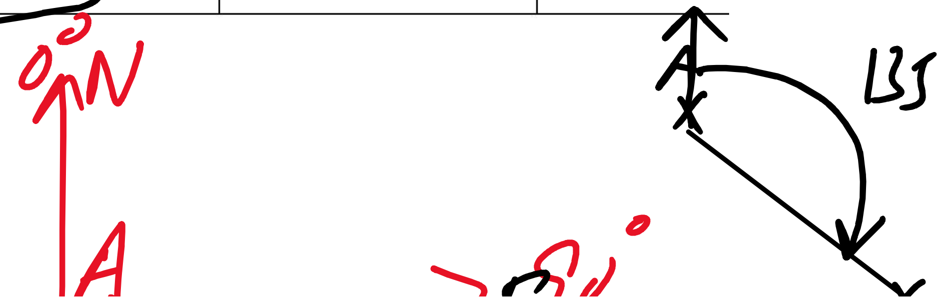
13. A ship sails on a three-figure bearing of 075°. In which direction is the ship heading?

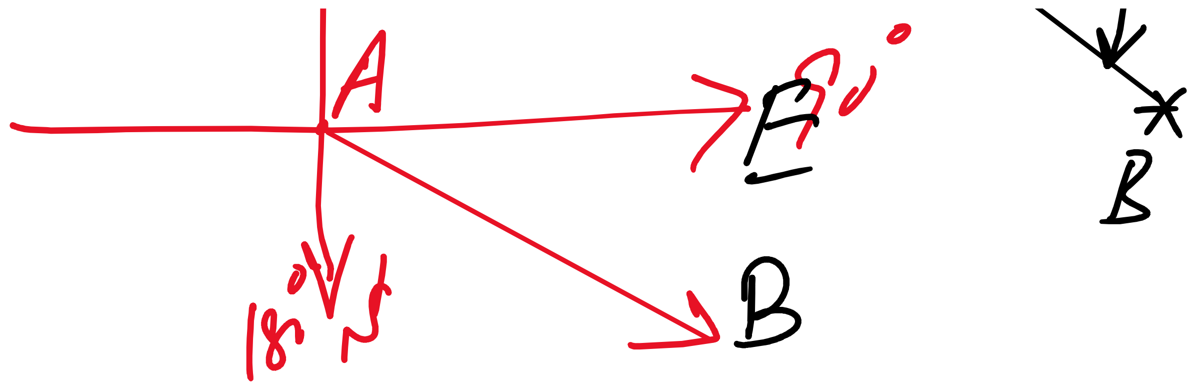
A) North	B) East	C) North-East	D) South-East



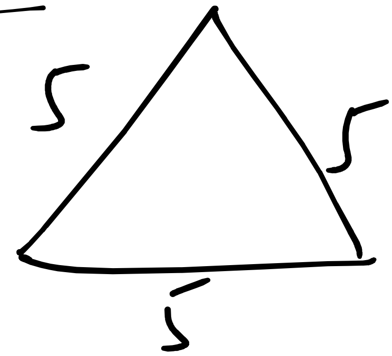
14. If the bearing of B from A is 135°, what direction is it?

A) North-West	B) South-East	C) South-West	D) North-East



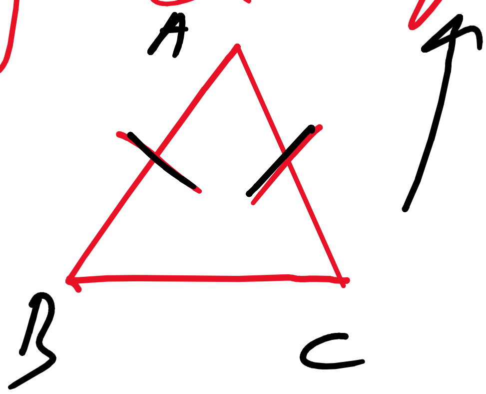


Equilateral △



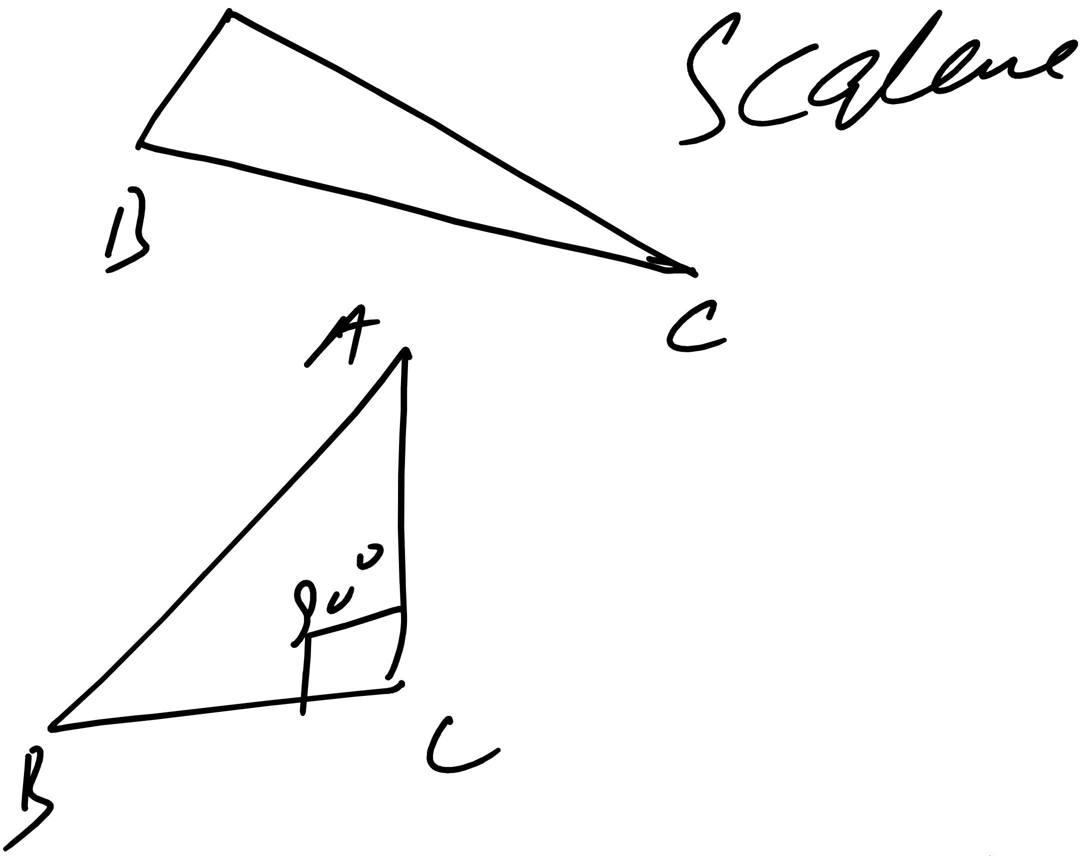
Isosceles △

Having 2 Equal sides



$AB = AC$

A $\quad AB \neq BC \neq AC$



Acute Angle $< 90°$

Obtuse Angle $> 90°$

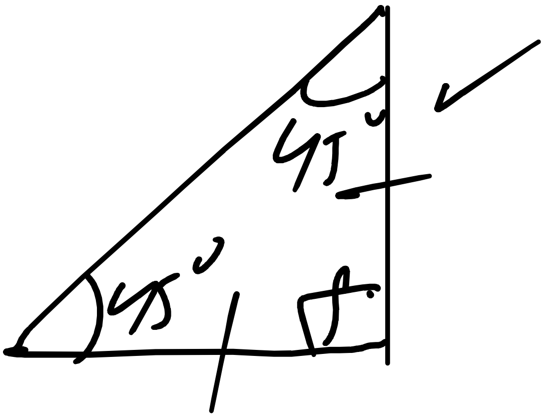
Right Angle $= 90°$

$\theta + \alpha = 90°$

→ Complementary Angle

↳ Complementary

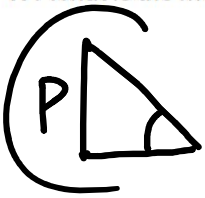
$\theta + \beta = 180°$ supplementary Angle

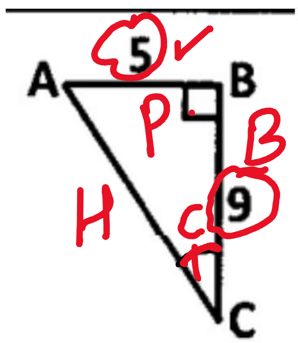


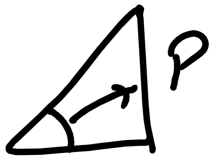
16. Which of the following identities is correct?

A) $\sin^2\theta + \cos^2\theta = 2$	B) $\sin^2\theta + \cos^2\theta = 1$	C) $\tan^2\theta = 1 + \sec^2\theta$	D) $\cos^2\theta - \sin^2\theta = 0$

17. What is the measure of angle C in the given figure?







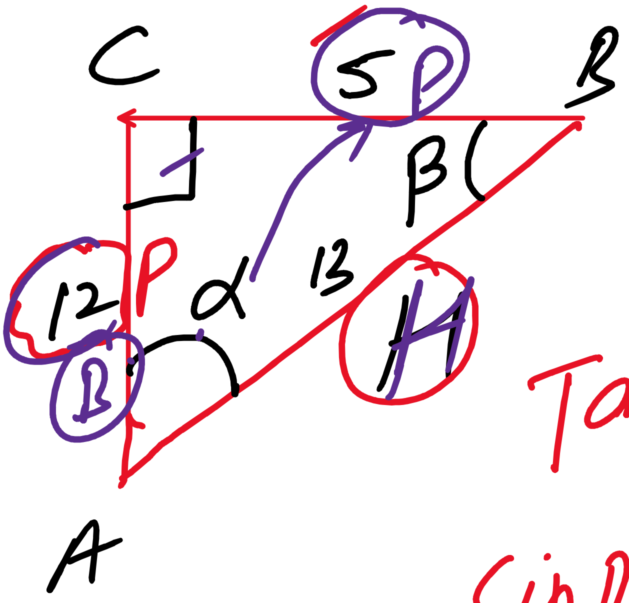
A) 29.1°	B) 60.94°	C) 56.25°	D) 33.75°

$\tan C = \frac{5}{9} \Rightarrow C = \tan^{-1}\left(\frac{5}{9}\right)$

$\tan C = \frac{5}{9} \Rightarrow C = \ldots$

γ

θ



$\cos\alpha = \frac{12}{13}$

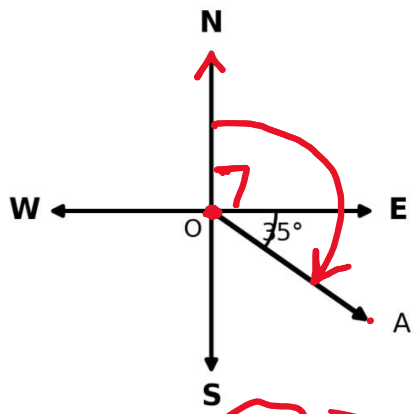
$\sin\alpha = \frac{5}{13}$

$\tan\beta = \frac{12}{5}$

$\sin\beta = \frac{12}{13}$

18. The bearing of A from O is:

$90^\circ + 35 = 125$



A) 035°	B) 055°	C) 125°	D) 305°

19. If an automobile travels on a bearing of 070° from city A to city B, then the bearing from city B to city A is:

A) 070°	B) 110°	C) 250°	D) 290°

True Bearing

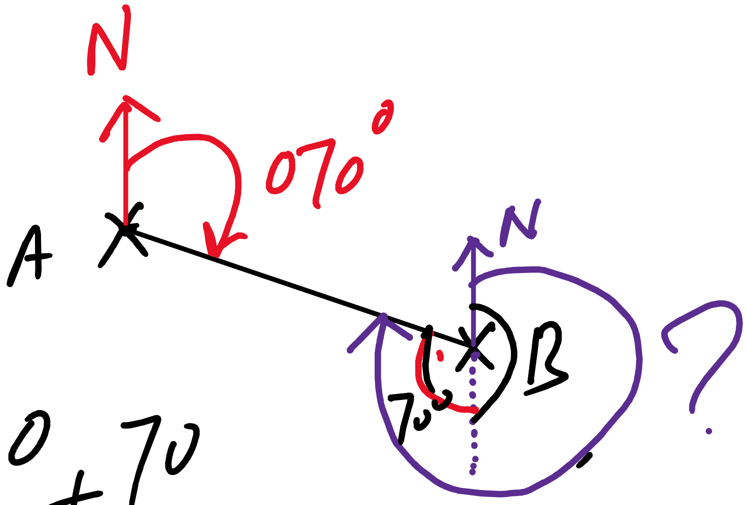
$A \rightarrow B$ 070° Give

$B \rightarrow A$

$70 + 180^\circ = 250^\circ$

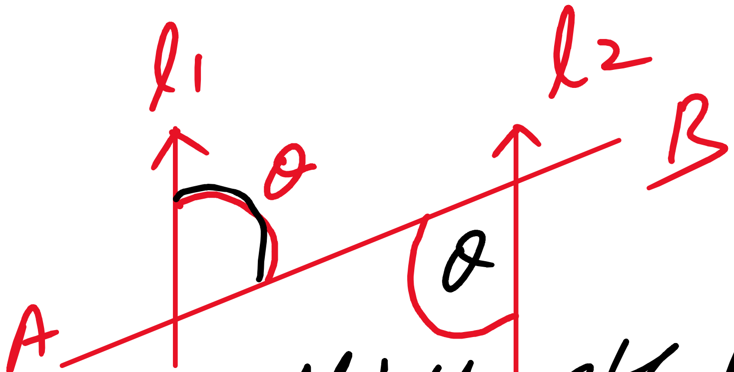
$B \rightarrow A$ 70 + 180° = 250

↳ Reverse Bearing



180° + 70

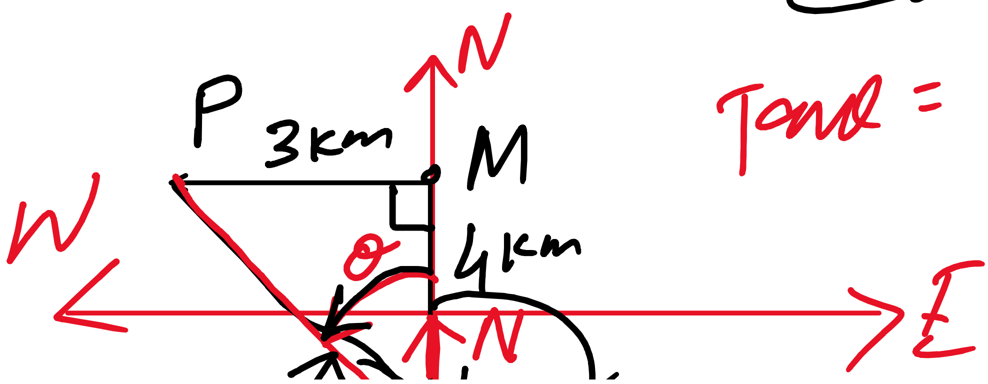
Transversal



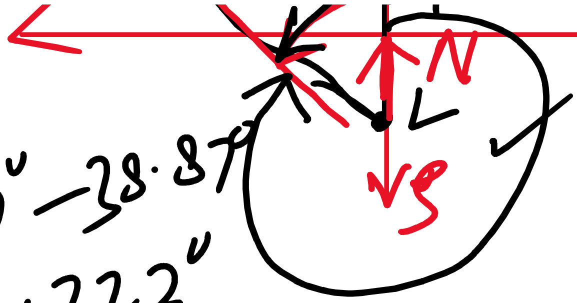
Alternate ∠

20. A boat sails 4 km north from L to M, then 3 km west from M to P. What is the bearing of P from L, to the nearest degree?

A) 037°	B) 053°	C) 217°	D) 323°



$\tan\theta = \frac{3}{4}$



$= 360° - 38.87°$

$\approx 323°$

$\theta = \tan^{-1}\left(\frac{3}{4}\right)$

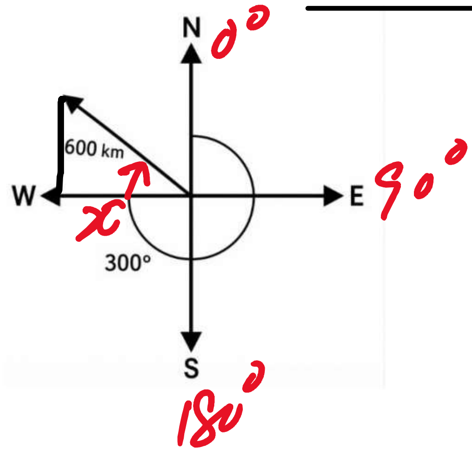
$\approx 38.87°$

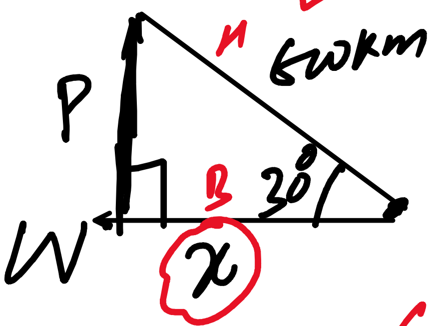
SECTION B - Short Questions

Q2. Answer all 20 short questions. Each question carries 4 marks. (20 × 4 = 80 marks)

Q2(1). A plane travels 600 km on a bearing of 300°. How far west did the plane travel? **[4 marks]**

Sol.





$$\cos 30° = \frac{x}{600\text{km}}$$

$$x = 600\text{km} \times \cos 30°$$

$$x = 600\text{ km} \times \frac{\sqrt{3}}{2}$$

$$\boxed{x \approx 520\text{km}}$$

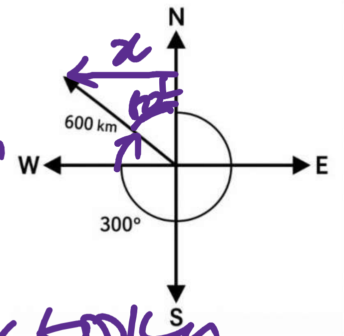
Method #2

SECTION B - Short Questions

Q2. Answer all 20 short questions. Each question carries 4 marks. (20 × 4 = 80 marks)

Q2(1). A plane travels 600 km on a bearing of 300°. How far west did the plane travel? **[4 marks]**

$$\sin 60 = \frac{x}{600km}$$

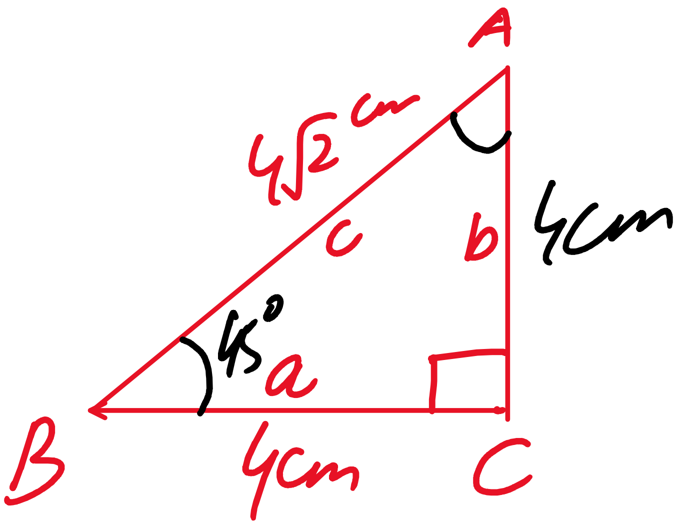


$$x = \sin 60 \times 600km$$

$$= \frac{\sqrt{3}}{2} \times 600km \doteqdot 520 \text{ km}$$

Q2(2). Solve the right-angled triangle ΔABC, given ∠C = 90°, a = 4 cm, and c = 4√2 cm. **[4 marks]**

Sol



By Pythagoras Theorem,

$$c^2 = a^2 + b^2$$

$$\Rightarrow b^2 = c^2 - a^2$$

$$b^2 = (4\sqrt{2})^2 - (4)^2$$

$$\Rightarrow b = \sqrt{16 \times 2 - 16}$$

$$b = \sqrt{32 - 16} = \sqrt{16}$$

$$\boxed{b = 4\,cm}$$

$\because$ $\tan B = \frac{4}{4} = 1$

$$B = \tan^{-1}(1) = 45^\circ$$

$$\boxed{B = 45^\circ}$$

$\because$ $A + B + C = 180^\circ$

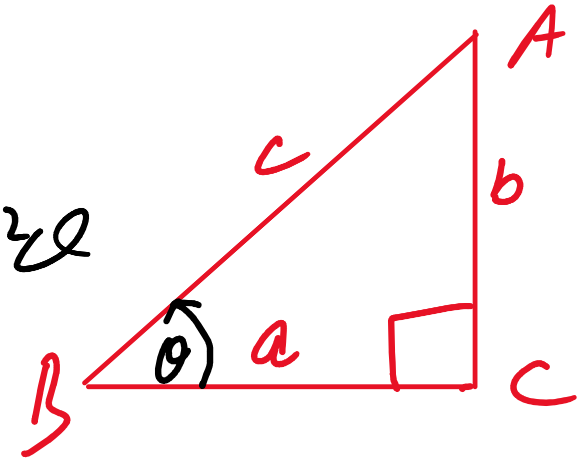
$$A = 180^\circ - B - C$$

$A = 180^\circ - 45^\circ - 90^\circ$

$\boxed{A = 45^\circ}$

Q2(3). Verify that $\sin^2\theta + \cos^2\theta = 1$. **[4 marks]**

Sol: Let $\triangle ABC$ is a Right Angled Triangle



L.H.S = $\sin^2\theta + \cos^2\theta$

$$= \left(\frac{b}{c}\right)^2 + \left(\frac{a}{c}\right)^2$$

$$= \frac{b^2}{c^2} + \frac{a^2}{c^2} = \frac{a^2+b^2}{c^2}$$

$$= \frac{\cancel{c^2}}{\cancel{c^2}} \quad \left(\because \text{As } c^2 = a^2+b^2 \text{ by Pythagoras theorem}\right)$$

$= \frac{\;}{c^2}$ (by Pythagoras Theorem)

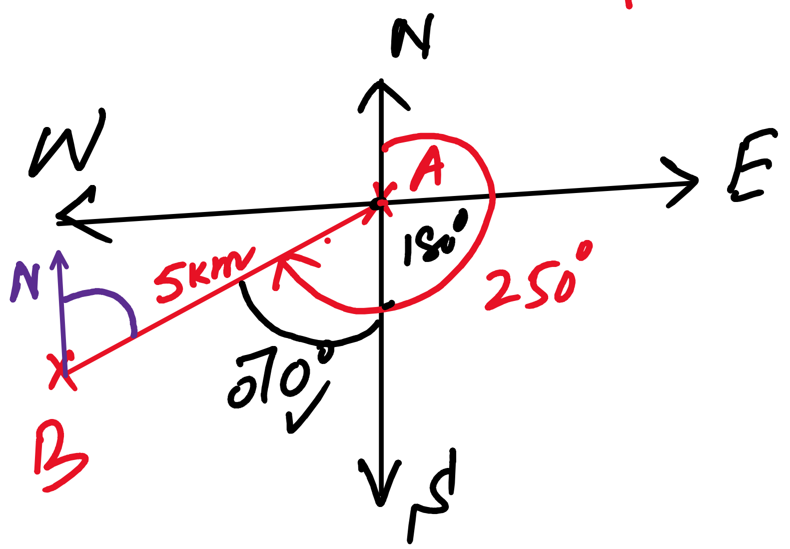
$= 1 = RHS$

Thus $\sin^2\theta + \cos^2\theta = 1$

prove

Q2(4). Two boats A and B are 5 km apart, and the bearing of B from A is 250°. Draw a diagram showing their relative positions, then find the bearing of A from B. **[4 marks]**

Sol:



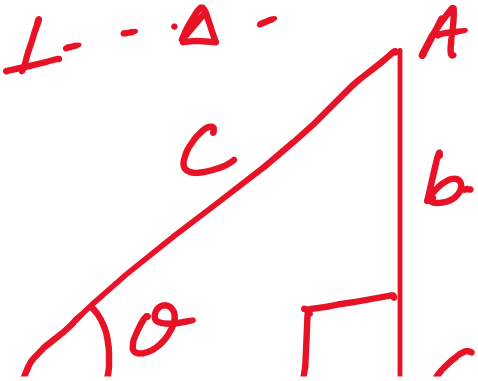
$180^\circ + 70^\circ = 250^\circ$

Bearing from B to A $= 250^\circ - 180^\circ$

$= 070^\circ$

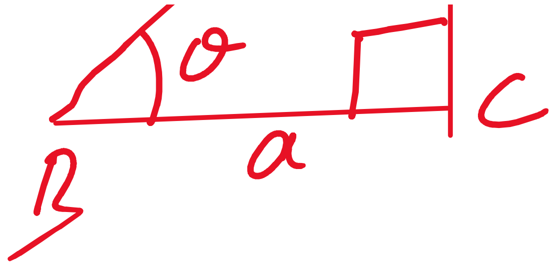
Q2(5). Show that $1 + \tan^2\theta = \sec^2\theta$. **[4 marks]**

Sol: ΔABC is $\perp$ ... Δ



$\tan\theta = \frac{b}{a}$

$\tan\theta = \frac{\ }{a}$



$\sec\theta = \frac{c}{a}$

$$LHS = 1 + \tan^2\theta$$

$$= 1 + \frac{b^2}{a^2} = \frac{a^2+b^2}{a^2}$$

$$= \frac{c^2}{a^2} \quad (\because c^2 = a^2+b^2 \text{ by Pythagoras Theorem})$$

$$= \left(\frac{c}{a}\right)^2$$

$$= \sec^2\theta = RHS$$

$$\Rightarrow LHS = RHS$$

Hence proved

Prove ~~that~~

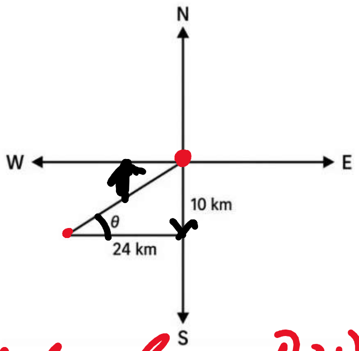
$\operatorname{cosec}^2\theta$

prove T

$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

$$\Rightarrow \operatorname{cosec}^2\theta - \cot^2\theta = 1$$

Q2(6). A cyclist travels 10 km due south and then 24 km due west. What simple bearing should the cyclist take to return directly to the starting point? **[4 marks]**



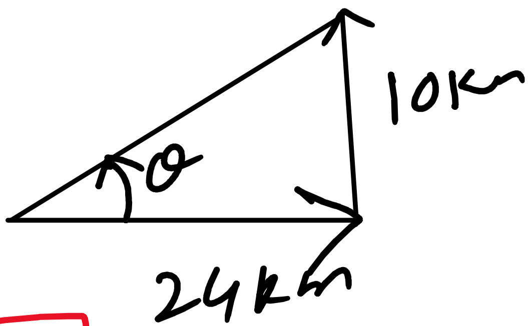
N/S; E/W

Sol:

To return to the starting point, he must travel 24 km ~~West~~ East and 10 km North

$$\tan\theta = \frac{10}{24}$$

$$\theta = \tan^{-1}\left(\frac{10}{24}\right)$$



22.6^0

$$\boxed{\theta = 22.62^\circ}$$

Simple Bearing $= 90^\circ - 22.62^\circ$

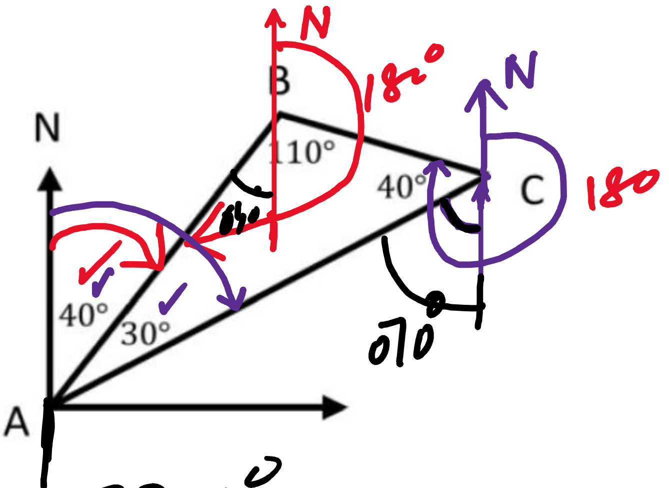
Reverse Bearing = $= 67.38^\circ$

$$= \boxed{N\ 67.38^\circ E}$$

AB

Q2(7). The figure shows the positions of A, B and C. State the bearing of: (i) B from A, (ii) A from B, (iii) B from C, and (iv) C from B. **[4 marks]**

Sol.



i) 040°

ii) $040^\circ + 180^\circ = 220^\circ$

iii) $180^\circ + 070^\circ + 040^\circ$

$= 290^\circ$

iv) $290^\circ - 180^\circ$

$$= 110^\circ$$

Q2(8). Prove that 1/(1 + cos x) + 1/(1 - cos x) = 2 + 2cot^{2}x. **[4 marks]**

$$L.H.S = \frac{1}{1+\cos x} + \frac{1}{1-\cos x}$$

$$= \frac{1-\cancel{\cos x}+1+\cancel{\cos x}}{(1+\cos x)(1-\cos x)}$$

$$= \frac{2}{1-\cos^2 x} = \frac{2}{\sin^2 x}$$

$$= 2\operatorname{cosec}^2 x$$

$$= 2(1+\cot^2 x)$$

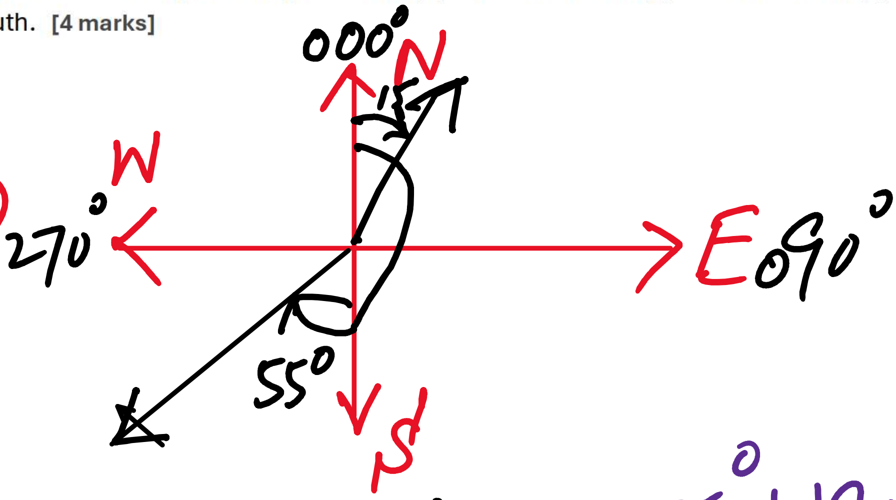
$$= 2 + 2\cot^2 x$$

$$= R.H.S$$

$L \cdot H \cdot S = R \cdot H \cdot S$

Q2(9). Write each in three-digit bearing notation: (a) South 55° West, (b) North 15° East, (c) North 110° West, (d) West 35° South. **[4 marks]**

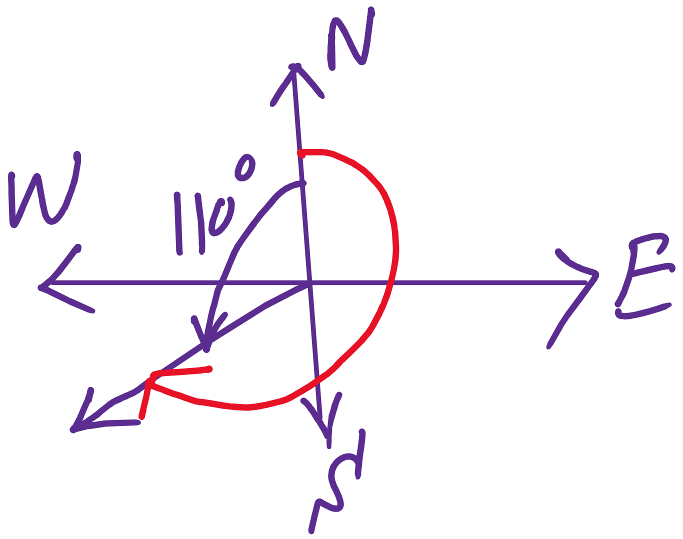
a)



$180° = 055° + 180° = 235°$

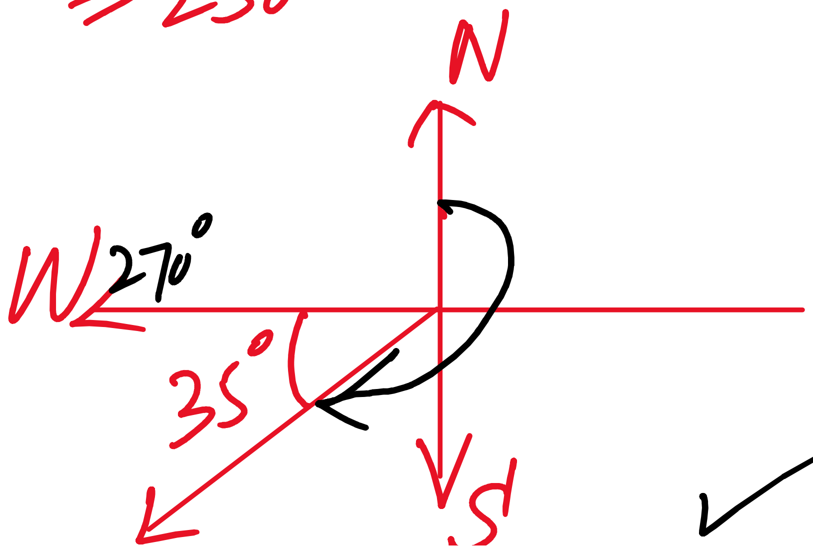
b) $015°$

c)



$360° - 110°$
$= 250°$

d)



S

$270^\circ - 35^\circ = 235^\circ$ Ar

Q2(10). Prove that sinθ/(1 - cosθ) - (1 + cosθ)/sinθ = 0. **[4 marks]**

$$\text{L.H.S} = \frac{\sin\theta}{1-\cos\theta} - \frac{1+\cos\theta}{\sin\theta}$$

$$= \frac{\sin^2\theta - (1+\cos\theta)(1-\cos\theta)}{\sin\theta(1-\cos\theta)}$$

$$= \frac{\sin^2\theta - (1-\cos^2\theta)}{\sin\theta(1-\cos\theta)}$$

$$= \frac{\sin^2\theta - \sin^2\theta}{\sin\theta(1-\cos\theta)} \quad \left(\because \sin^2\theta + \cos^2\theta = 1\right)$$

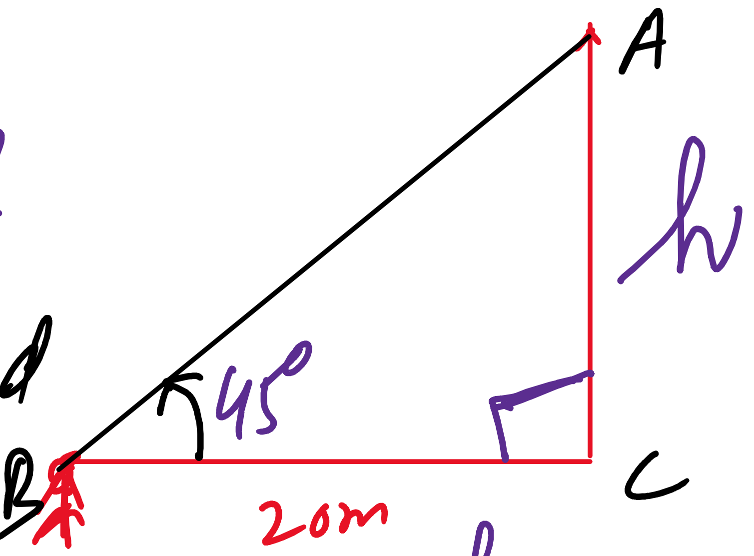
$$= \frac{0}{\sin\theta(1-\cos\theta)} = 0 = \text{RHS}$$

$$= \frac{\quad}{\sin\theta(1-\cos\theta)} =$$

Q2(11). A person is standing 20 m away from a tree. The angle of elevation from the person's eye level to the top of the tree is 45°. (a) Find the height of the tree. (b) If the person moves 10 m closer, find the new angle of elevation. **[4 marks]**

Sol:

a) $h = ?$

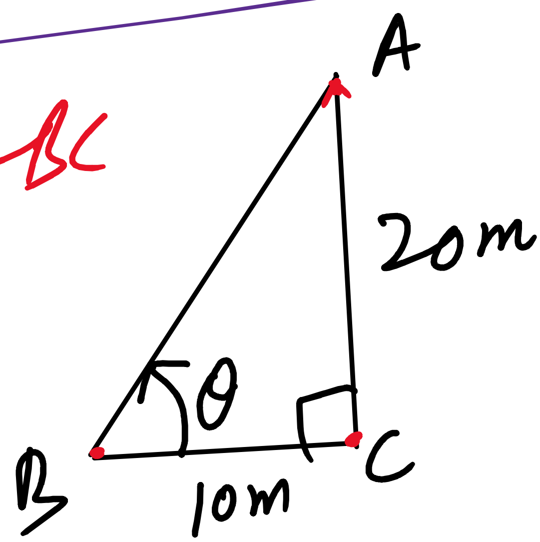


In Rt. Angled $\triangle ABC$, B

$$\tan 45^\circ = \frac{h}{20\,m} = 1.$$

$$\Rightarrow \boxed{h = 20\,m}$$

b) In Rt. $\triangle ABC$



$$\tan\theta = \frac{20\,m}{10\,m}$$

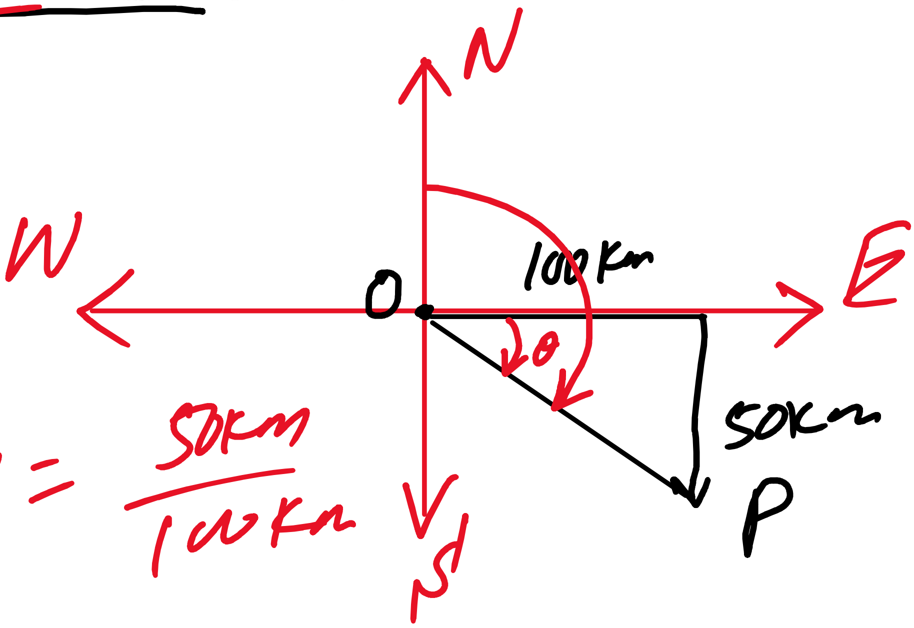
$$\theta = \tan^{-1}(2)$$

$\theta = Tan^{-1}(2)$

$\theta \approx 63.43^\circ$

Q2(12). A person travels 100 km due East, then 50 km due South. Find the direction and bearing of the final position from the starting point. **[4 marks]**

Sol



$\tan\theta = \frac{50km}{100km}$

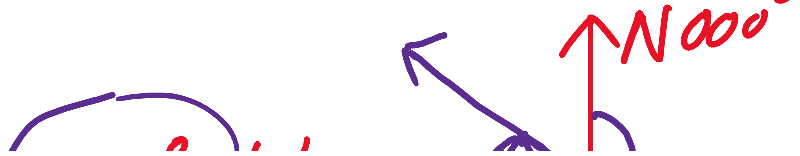
$\theta = Tan^{-1}(\frac{1}{2}) \approx 26.6^\circ \approx 027^\circ$

True Bearing $= 090^\circ + 027^\circ = 117^\circ$

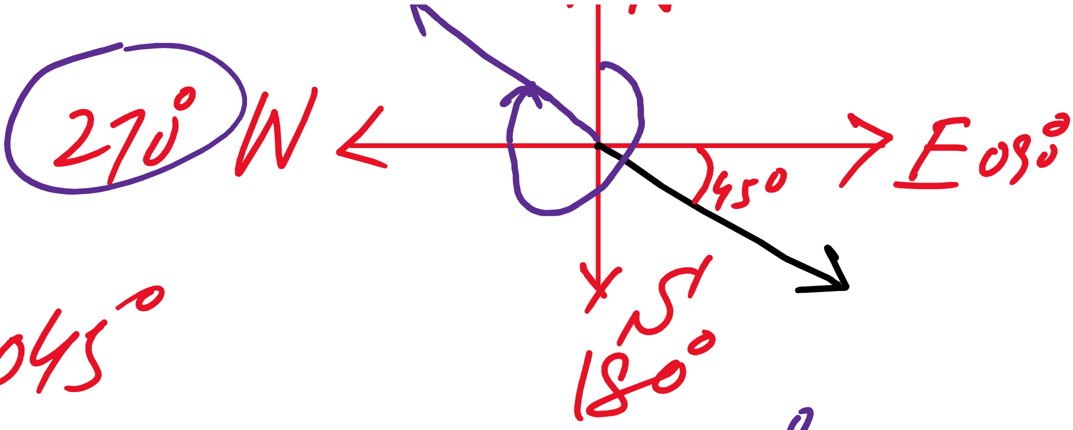
Final Direction = South-East

Q2(14). State the three-digit bearing for a ship heading: (a) East, (b) South-East, (c) West, (d) North-West. **[4 marks]**

a) 090°



a) 090°



b) $090^\circ + 045^\circ$

$= 135^\circ$

c) $270^\circ -$

d) $270^\circ + 045^\circ = 315^\circ$

Q2(15). Prove that sinθ/(1 - cosθ) = cosecθ + cotθ. **[4 marks]**

$$L.H.S = \frac{\sin\theta}{1-\cos\theta} \times \frac{1+\cos\theta}{1+\cos\theta}$$

$$= \frac{\sin\theta(1+\cos\theta)}{1-\cos^2\theta}$$

$$= \frac{\cancel{\sin\theta}(1+\cos\theta)}{\sin^2\theta}$$

$$= \frac{1 + \cos\alpha}{\sin\alpha}$$

$$= \frac{1}{\sin\alpha} + \frac{\cos\alpha}{\sin\alpha}$$

$$= \operatorname{cosec}\alpha + \cot\theta.$$

$= R.H.S$

$L.H.S = R.H.S$

Hence proved